

## Title

Evidence Before Presentation: A Verdict-Producing Trust Pipeline for News and Multimodal Social Content, with Independence-Aware Corroboration

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### 1. Abstract

Systems that judge online information almost always score how an article is written. That is the wrong quantity, and it fails in a specific and dangerous direction: everything such a system measures — the calm register, the named sources, the absence of clickbait — is under the control of whoever wrote the text. A fabricated story composed in plain newsroom style therefore scores well, and the more competent the fabrication, the better it scores. We measured this on our own earlier pipeline: invented articles averaged 7.3 out of 10, and four of six deceptive items were presented to the reader as credible.

The cause was an inverted treatment of missing evidence. When no independent outlet could be found reporting a story, the cross-source check withdrew and its weight was redistributed across the remaining checks — so an absence of corroboration increased the influence of exactly those signals a fabricator controls. This work rebuilds the system on the opposite principle: **presentation may only lower trust; evidence is what raises it**. The five content checks produce a presentation score, the evidence channel produces a verdict, and the final score is the lesser of the presentation score and the ceiling that verdict permits. A story nobody else reports cannot be called trustworthy however well it is written.

The evidence channel contributes two further mechanisms. First, corroboration is counted in **independent sources rather than articles**: a curated ownership model, agency-syndication detection and near-duplicate collapse reduce eight retrieved articles running one wire story to the single source they actually represent, and every merge is reported to the reader with its reason. Second, a **deterministic decision procedure** assigns one of seven verdicts — corroborated, likely true, unverified, commentary, satire, likely false, false — naming the rule that fired, the grounds, and a confidence derived only from the quantity and quality of independent evidence. The system states what it concluded and why, and it says *unverified* rather than guessing when nothing supports a claim.

Two extensions follow from the same architecture. Language is identified locally across fifteen languages and nine scripts, and claims are searched for corroboration in both the original language and English, so a Tamil or Hindi report can be confirmed by English coverage and vice versa. Non-article content is normalised into the same representation: a YouTube video is analysed from its transcript, a screenshot through optical character recognition, and a forwarded message as pasted text — each carrying a provenance record, because an anonymous account is not a publisher and must not inherit a publisher's neutral rating.

Critically, every decisive step — independence, ceiling, verdict, calibration, language identification — is deterministic local code. The language model is confined to extraction and stance judgment over retrieved text; it is a sensor, not the judge. The project's test suite and adversarial benchmark both run with every external key removed, so the contribution is separable from the borrowed model and the borrowed news feed.

**Keywords** — misinformation assessment, evidence-grounded verification, source independence, verdict explanation, multilingual fact verification, multimodal content credibility, deterministic decision procedures

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## 2. Base Paper

**S. Amri, H.-C. Mputu Boleilanga, and E. Aïmeur, "ExFake: Towards an Explainable Fake News Detection Based on Content and Social Context Information," arXiv:2311.10784, Nov. 2023.**

**What the base paper does.** ExFake judges the veracity of an online post by fusing three families of signal — the content of the post, social-context information (the credibility and historical behaviour of the users circulating it), and evidence from trusted external entities such as fact-checking websites and named-entity records — and couples the decision to an explainable-AI assistant intended to help an ordinary social-network user reason about doubtful information.

**Why this paper.** It matches this project on objective (help a non-expert reader judge a post rather than classify it), on method (fusion of heterogeneous signal families rather than one classifier), on external grounding, and on audience. Its social-context dependency is also the hinge of our extension: we replace signals about *who is sharing a post* — unavailable at first sight, which is when a reader needs an answer — with signals about *who published it and whether anyone independent confirms it*.

### Alternatives considered and rejected as the base.

- *VeraCT Scan* [3] (2024) searches for corroborating and conflicting reports and justifies its verdict — the nearest neighbour to our verification channel. Not chosen because it already occupies that ground; our remaining contributions read better as improvements on a full reader-facing assessment.
- *TELLER* [2] (Findings of ACL 2024) explains through logic rules rather than retrieved evidence, so its explanation is a different object from ours.
- *TRUST-VL* [6] (EMNLP 2025) is the strongest recent multimodal explainable system, but it is a vision-language model trained on a 198K-sample corpus; adopting it as the base would frame this project as an under-resourced attempt at the same thing, whereas our contribution is architectural. It is the state of the art against which our multimodal work is positioned, not the paper being extended.

**Currency.** Every comparative reference in Section 8 is from 2023 or later. The two pre-2020 entries are cited only as the provenance of standard techniques and are listed separately.

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## 3. The Defect This Work Addresses

Prior systems, including the base paper and our own earlier pipeline, combine their signals additively. That is sound only while every signal is independent evidence. It is not:

| Signal          | Controlled by the author?    |
|-----------------|------------------------------|
| Headline style  | Yes — write a plain headline |
| Neutral wording | Yes — write calmly           |

| Signal                              | Controlled by the author?                                   |
|-------------------------------------|-------------------------------------------------------------|
| Absence of persuasion techniques    | Yes — omit them                                             |
| Named sources and attributed quotes | <b>Yes — invent them</b>                                    |
| Publisher reputation                | Partly — publish on a new domain, receive a neutral default |
| Independent corroboration           | <b>No</b>                                                   |

Only the last row resists the author. An additive model therefore lets five author-controlled signals outvote the one that cannot be faked — and in the earlier design, when that one signal found nothing, its weight was handed to the other five. Measured on an adversarial set of invented articles written in ordinary newsroom register (Section 7.2), this produced a mean presentation score of 7.3/10 and presented four of six deceptive items as credible.

We treat this as a correctness defect rather than a tuning problem. No choice of weights repairs it, because the failure is in the direction of the inference: absence of corroboration was being read as absence of information, when for a newsworthy claim it is evidence in its own right.

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#### 4. Proposed System and Points of Novelty

All six claims below are implemented and under test in the submitted system.

**N1 · Separation of presentation from evidence, with a verifiability ceiling.** The five content factors yield a presentation score. The evidence channel yields a verdict, and each verdict class carries the highest trust score it permits: *false* 2.0, *likely false* 3.5, *satire* 4.0, *unverified* 4.9, *commentary* 6.4, *likely true* 8.0, *corroborated* 10. The reported score is min(presentation, ceiling). Because "unverified" caps below the "mostly fine" band, an uncorroborated story can never be described to a reader as fine. The report shows both numbers, so the distance between *reads well* and *is supported* is visible rather than averaged away.

*A necessary exception, stated in the rules:* an exclusive from an outlet with a strong factual record is not the same object as an anonymous site's uncorroborated claim, though both are unverified. Where the publisher's record is 8.5/10 or better, the ceiling relaxes to 6.4 — still short of "trustworthy", because the claim remains unconfirmed.

**N2 · Independence-aware corroboration.** Counting articles overstates corroboration, often by an order of magnitude. Three deterministic mechanisms reduce retrieved coverage to the sources it actually represents: a curated ownership model of 70 media groups (so Vox and The Verge, or Wired and Ars Technica, count once), detection of agency syndication from dateline and attribution credits (so eight papers running one Reuters story count once), and near-duplicate headline collapse for unattributed reprints. State-controlled outlets of the same state are merged for the same reason. Every merge is reported with its justification. No API supplies this information; the model is the project's own.

**N3 · A dispositive verdict with stated grounds.** Nine ordered rules assign one of seven verdict classes. Each returns the rule identifier that fired, the grounds in plain language, and a confidence derived from the count and quality of independent evidence — never from the model's own expressed certainty, which is not measurable. Two independent contradictions are the only route to a flat *false*; one yields *likely false*. Confidence is capped below 1: the system does not claim certainty.

**N4 · Cross-lingual corroboration.** Language is identified locally by script and function-word frequency across fifteen languages and nine scripts, with no model call and no network. Prompts instruct the model to analyse in the article's language and to quote spans without translating them, so highlighted evidence still occurs in the article. Claims are then searched in the original language *and* in English. This is what makes verification work outside English at all: an English-only search finds nothing for a Tamil report and concludes, wrongly, that no other outlet covers the story.

**N5 · Provenance-aware multimodal ingestion.** A YouTube link is analysed from its transcript — selecting the caption track in the language actually spoken rather than one of the machine translations — a screenshot through OCR, and a forwarded message as text. Each carries a provenance record. An account that is a known newsroom's channel inherits that newsroom's record; an anonymous account does not inherit the neutral 5/10 that an unrated newspaper receives, because there is no masthead, correction policy or accountability behind it.

**N6 · The decisive logic is deterministic and local.** Independence, ceilings, verdict rules, calibration and language identification contain no model call. The language model extracts claims and judges stance over retrieved text; it is a sensor, not the judge. The test suite (57 tests) and the adversarial benchmark both run with every API key removed, which makes the claim checkable rather than rhetorical: the contribution is what remains when the borrowed components are taken away.

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## 5. What the System Does Not Claim

Stated explicitly, because overclaiming is the fastest way to lose a review.

- **It cannot certify truth.** *Corroborated* means independent outlets report the same facts, which is the strongest evidence available without primary investigation. It is not proof.
- **It abstains, and says so.** *Unverified* is a first-class outcome, not a failure. Most claims outside major news land there, and reporting that honestly is the design, not a gap in it.
- **It does not perform synthetic-media forensics.** Deepfake and generated-image detection are a separate research problem. The architecture admits such a detector as an additional factor whenever a reliable one exists.
- **Its ownership model is hand-curated** and therefore incomplete. Unlisted outlets are never merged, which under-counts dependence rather than inventing independence.

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## 6. Evaluation

| Claim                                         | How it is evidenced                                                                                    |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------|
| The pipeline resists well-written fabrication | Adversarial benchmark (Section 7.2): legacy and current scoring computed from identical factor outputs |
| Independent counting is correct               | Unit tests over ownership, syndication, duplication and state-media cases                              |
| Verdicts are deterministic and reasoned       | Every verdict asserts its rule and grounds; tested across all seven classes                            |
| Language identification works                 | Fifteen languages, nine scripts, tested per language                                                   |

| Claim                                       | How it is evidenced                                                                                            |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| The contribution is separable from the APIs | Whole suite and benchmark execute with all keys removed                                                        |
| Decomposition outperforms a single prompt   | Leakage-controlled ISOT benchmark against a single-prompt baseline over the same model                         |
| Confidence is honest                        | Ten-bin expected calibration error, raw and Platt-scaled, fitted on validation and reported on a held-out half |

## 7.1 Status of the ISOT result

The earlier 94.7% / 82.0% comparison and the  $0.277 \rightarrow 0.073$  calibration figures were measured on a four-factor configuration using a language model since withdrawn by its provider. They are reported as the measurement that motivated the architecture. A re-measurement of the present configuration is outstanding.

ISOT also cannot test the defect in Section 3: its fabricated articles are mostly badly written, so a presentation-scoring system performs well on it while remaining blind to a competent fabrication. That is precisely why the adversarial set exists.

## 7.2 Adversarial benchmark

Ten hand-written items — five fabrications in plain newsroom register, one high-impact fabrication with manipulation signals, two genuine reports, one satirical piece and one commentary — each scored twice from identical factor outputs: once under the legacy additive rule with abstention redistribution, and once under the current rule. Run with the language model disabled, so the figures reflect the deterministic path alone:

| Measure                                                   | Legacy   | Current                 |
|-----------------------------------------------------------|----------|-------------------------|
| Deceptive items presented as credible (score $\geq 5.5$ ) | 4 / 6    | 0 / 6                   |
| Verdict matched expectation on deceptive items            | —        | 6 / 6                   |
| Genuine articles wrongly condemned                        | —        | 0 / 2                   |
| Mean presentation score of the fabrications               | 7.3 / 10 | (unchanged — the point) |

The last row is the finding worth stating plainly: the fabrications still *read* as trustworthy, and always will. What changed is that reading well no longer determines the answer.

## 8. References

### Primary and comparative work (2023–2026)

- [1] S. Amri, H.-C. Mputu Boleilanga, and E. Aïmeur, "ExFake: Towards an Explainable Fake News Detection Based on Content and Social Context Information," arXiv:2311.10784, 2023. **(Base paper)**

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- [3] C. Niu, Y. Guan, Y. Wu, J. Zhu, J. Song, R. Zhong, K. Zhu, S. Xu, S. Diao, and T. Zhang, "VeraCT Scan: Retrieval-Augmented Fake News Detection with Justifiable Reasoning," arXiv:2406.10289, 2024.
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- [9] D. Yang, "Calibrated Selective Fact-Checking via Evidence Chain Evaluation," arXiv:2607.18240, 2026.

## Dataset

- [10] H. Ahmed, I. Traore, and S. Saad, "Detection of Online Fake News Using N-Gram Analysis and Machine Learning Techniques," in *Proc. Int. Conf. Intelligent, Secure, and Dependable Systems in Distributed and Cloud Environments*, 2017, pp. 127–138. (Source of the ISOT corpus used for evaluation, not comparative work.)

## Methodological foundations

Cited for the provenance of standard techniques rather than as contemporary comparison.

- [11] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On Calibration of Modern Neural Networks," in *Proc. 34th Int. Conf. Machine Learning (ICML)*, 2017, pp. 1321–1330.
- [12] J. Platt, "Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods," in *Advances in Large Margin Classifiers*, MIT Press, 1999, pp. 61–74.

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## Appendix A — Short Abstract (150 words)

Systems that judge news score how an article is written, which is the wrong quantity: every such signal is controlled by whoever wrote the text, so a competent fabrication scores well. Our earlier pipeline rated invented articles 7.3 out of 10 and presented four of six deceptive items as credible, because when no corroborating coverage was found the check withdrew and handed its weight to the signals a fabricator controls. This work inverts that: presentation may only lower trust, evidence alone may raise it, and the score is capped by what independent reporting supports. Corroboration is counted in independent sources rather than articles, collapsing common ownership and wire syndication; a deterministic rule set then issues one of seven verdicts with its grounds and confidence. Language is identified across fifteen languages and claims searched across language boundaries; video, screenshot and forwarded text are ingested with provenance. Under the new rule, zero of six deceptive items read as credible.